

Clavra ProdPilot™

AI-Powered Manufacturing ERP — Project Summary | v1.1 | June 2026

1. Brief Summary

Clavra ProdPilot™ is a full-stack, AI-driven ERP system purpose-built for garment manufacturing. It replaces WhatsApp and Excel-based workflows with a real-time dashboard, an intelligent AI Copilot (GPT-4o + Ollama), a RAG-powered Knowledge Base, and an AI Vision module for automated quality inspection — enabling factory managers to query production status, monitor shipments, manage inventory, retrieve SOPs, and detect fabric defects using natural language or voice.

2. Problem Statement

Small and mid-size garment factories in South & South-East Asia rely on WhatsApp messages and Excel sheets to manage production orders, shipments, and raw material inventory. Quality control is done manually with paper checklists, and defect images are shared over chat with no traceability. Clavra ProdPilot™ delivers an affordable, AI-first ERP where workers simply ask questions in plain language and photograph defects directly in the chat interface.

3. Key Features

Feature	Description
AI Copilot (18+ Intents)	GPT-4o intent classification with keyword + Ollama fallbacks; voice input via Web Speech API
AI Vision — Defect Analysis	Upload or photograph garments/fabric; GPT-4o Vision (primary) or Ollama moondream/llava (local) detects defects, reads labels, inspects equipment. Returns severity, findings, and recommendations. Results stored in vision_analyses table with full audit trail.
Production Management	Track orders, progress (produced vs target), defect qty, delivery dates with inline editing
Shipment Tracking	Real-time status lifecycle linked to production orders
RAG Knowledge Base	Upload PDFs/DOCX/XLSX SOPs; semantic vector search answers with citations

Inventory Management	Material stock, reorder alerts, supplier info
Production Lines	Line A/B/C/D assignment, status, and efficiency tracking
Quality Control	Defect analysis, QC records, inspection reports
Analytics Dashboard	KPI cards, order trend charts, inventory analytics
Multi-tenant RBAC	Org-scoped data with Admin/Manager/Operator roles
WebSocket Streaming	Real-time AI responses streamed token-by-token with intent badges

4. Technology Stack

Layer	Technologies
Backend	Python 3.11+, FastAPI, SQLAlchemy 2.0 (async), Alembic, Uvicorn
Database	PostgreSQL 16 + pgvector (1536-dim vector embeddings)
Frontend	React 18, TypeScript, Vite 5, Tailwind CSS v4, Zustand
AI — Language	OpenAI GPT-4o (intent + RAG answers), Ollama llama3.1:8b (local fallback)
AI — Vision	OpenAI GPT-4o Vision (primary), Ollama moondream:latest (local), Ollama llava (fallback)
AI — Embeddings	text-embedding-3-small (1536d, cloud), nomic-embed-text (local fallback)
Auth	JWT (python-jose), bcrypt, refresh token rotation
Voice	Web Speech API (browser-native STT + TTS)
DevOps	Docker Compose, Nginx reverse proxy, Alembic migrations

5. Architecture Overview

Three-tier architecture:

Browser (React/TS) ↔ FastAPI REST + WebSocket ↔ PostgreSQL + pgvector

AI intent pipeline (three stages):

Stage 1 — detect_intent(): GPT-4o (8s timeout) → keyword rules → Ollama fallback

Stage 2 — route_intent(): 4 override guards correct misclassifications → dispatch

Stage 3 — Handler: DB query / RAG / SQL generator / Vision service

WebSocket sends intent badge immediately after Stage 1, then streams handler tokens.

6. AI Vision Module

The AI Vision module allows shop-floor workers to photograph garments, fabric rolls, labels, or production line equipment directly inside the AI Copilot chat. The image is analyzed automatically and a structured quality report is returned with findings, severity rating, and actionable recommendations.

6.1 Supported Analysis Types

Analysis Type	Description
defect_detection	Identifies visible defects on fabric or finished garments: holes, stains, seam failures, colour inconsistencies, pilling, snags
label_reading	Reads and validates care labels, size labels, country-of-origin labels for compliance checks
line_photo	Assesses overall production line or workstation — worker ergonomics, machine setup, material placement
equipment_check	Inspects sewing machines, cutting equipment, or pressing irons for visible wear or maintenance needs

6.2 AI Model Fallback Chain

Priority	Model	Behaviour
1st	GPT-4o Vision (OpenAI)	Returns structured JSON with findings, severity, defect_rate_estimate, confidence. Requires valid OpenAI API key with vision credits.
2nd	Ollama moondream:latest	Free local vision model (~1.6 GB). Runs offline. Returns parsed structured response. Pull with: ollama pull moondream
3rd	Ollama llava:latest	Secondary local fallback if moondream not installed. Larger model (~4 GB). Pull with: ollama pull llava

4th	Graceful fallback	Returns a user-friendly message with instructions to enable vision models. No crash — always returns a usable response.
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6.3 Vision Response Structure

Every vision analysis returns a standardised JSON object regardless of which model ran:

```
{
  "analysis_type":      "defect_detection | label_reading | line_photo |
equipment_check",
  "findings":           ["finding 1", "finding 2", ...],
  "severity":           "critical | major | minor | none",
  "recommendations":    ["action 1", "action 2", ...],
  "defect_rate_estimate": 0.05,
  "confidence":         0.92,
  "summary":           "Human-readable 2-sentence summary"
}
```

6.4 Frontend Integration

The AI Copilot chat includes an image upload button (camera icon) beside the message input. Workers tap the button, select or photograph an image, and the analysis result appears as a structured chat bubble alongside the conversation. The ImageUploadZone component accepts any image format (JPEG, PNG, WEBP) via the browser file picker.

6.5 Data Persistence

Column (vision_analyses table)	Description
id	Primary key
org_id	Organisation scope (multi-tenant)
user_id	Analyst who uploaded the image
image_filename	Original uploaded filename
analysis_type	defect_detection / label_reading / line_photo / equipment_check
findings	JSON array of individual findings
defect_rate	Estimated defect rate (0.0 – 1.0)
confidence	Model confidence score (0.0 – 1.0)
recommended_action	Concatenated recommendation text
summary	Human-readable summary
created_at	Timestamp of analysis

7. Development Access

Frontend: <http://localhost:5174>

Backend API: <http://localhost:8000>

API Docs: <http://localhost:8000/docs>

Admin Login: admin@clavra.com / Admin@123

Ollama: <http://localhost:11434>

Vision models: ollama pull moondream (local vision, ~1.6 GB)